

CS 334 Project Report

Written by Henry Lee and Nathan Laury

Project Title & Team Members

- Project Title(s):
 - Neural 3D Shape Generator
 - Shape Synthesis
- Team Members:
 - Nathan Laury (model training, generation)
 - Henry Lee (point cloud rendering, visualization)

Background & Motivation

Recent advances in machine learning have enabled new approaches to the field of automated 3D content generation, particularly the intersection of artificial intelligence and computer graphics. Generating 3D objects using neural networks has many applications, including modeling, simulation, game development, and procedural asset creation. The goal of this project is to explore how generative models produce 3D geometry and how these outputs can be integrated into a graphics pipeline with visualization support. Specifically, we developed a pipeline that generates point clouds from a neural network, converts them into 3D models using a surface reconstruction algorithm, and displays them in a real-time viewer. This project is motivated by the increasing integration of machine learning into graphics workflows, where it complements traditional rendering and geometry processing methods.

Related Work & Prior Art

Previous works in 3D shape generation have explored a variety of representations, including voxels, meshes, implicit fields, and point clouds. Among these, point cloud-based methods are widely used due to their simplicity and efficiency, and they are commonly paired with generative models such as variational autoencoders (VAEs) and generative adversarial networks (GANs). Additionally, geometry processing techniques such as surface reconstruction are often used to convert point clouds into mesh representations, with common algorithms like alpha shapes and Poisson reconstruction. Python libraries such as Open3D provide practical implementations of these methods and are used as a foundation for the mesh generation stage in this project. These prior approaches shaped the design of our generative model and the overall processing pipeline.

Technical Approach

The overall pipeline consists of three stages: generating a point cloud, converting it into a 3D mesh, and displaying it in an interactive viewer.. First, a dataset of 3D point clouds is used to train a variational autoencoder (VAE) that learns a latent representation of object geometry. The encoder aggregates point-wise features into a global latent representation, which enables the model to generate diverse yet structurally consistent shapes. After training, new shapes are generated by sampling random latent vectors and decoding them into point clouds. For this project, we chose chairs as a starting point to focus on the overall pipeline. The generated point clouds are then processed to improve quality, including normalization, outlier removal, and normal estimation. These processed points are converted into a 3D model using alpha shape surface reconstruction, producing a triangular mesh suitable for rendering. Finally, the reconstructed mesh is displayed in an interactive viewer that supports real-time rendering and

user controls like rotation and zoom. This integrates machine learning-based geometry generation with a graphics pipeline for visualization, meeting the project goal of combining AI-driven content generation with interactive graphics.

Timeline & Development Process

The project followed the planned development stages outlined in the proposal. Initial work focused on implementing a basic viewer capable of rendering point clouds and supporting camera interaction. The next phase involved training the VAE model and generating point cloud outputs. After model training, the focus shifted to improving visualization through surface reconstruction and mesh rendering. Additional work was required to refine point cloud preprocessing and stabilize mesh generation. Responsibilities were divided between machine learning and graphics components, with one team member focusing on model development and the other on rendering and visualization.

Current Results

The pipeline successfully generates 3D chairs as point clouds and converts them into meshes for visualization. The generated chairs showcase variation while maintaining recognizable structural features from the training data, demonstrating that the model learned meaningful shape representations. Mesh reconstruction improves the visual quality of the generated chairs, making them easier for viewers to interpret. The interactive visualization system allows users to rotate, zoom, and inspect generated shapes in real time.

The system includes four main interface stages: the landing page, generation screen, mesh construction screen, and final interactive viewer.

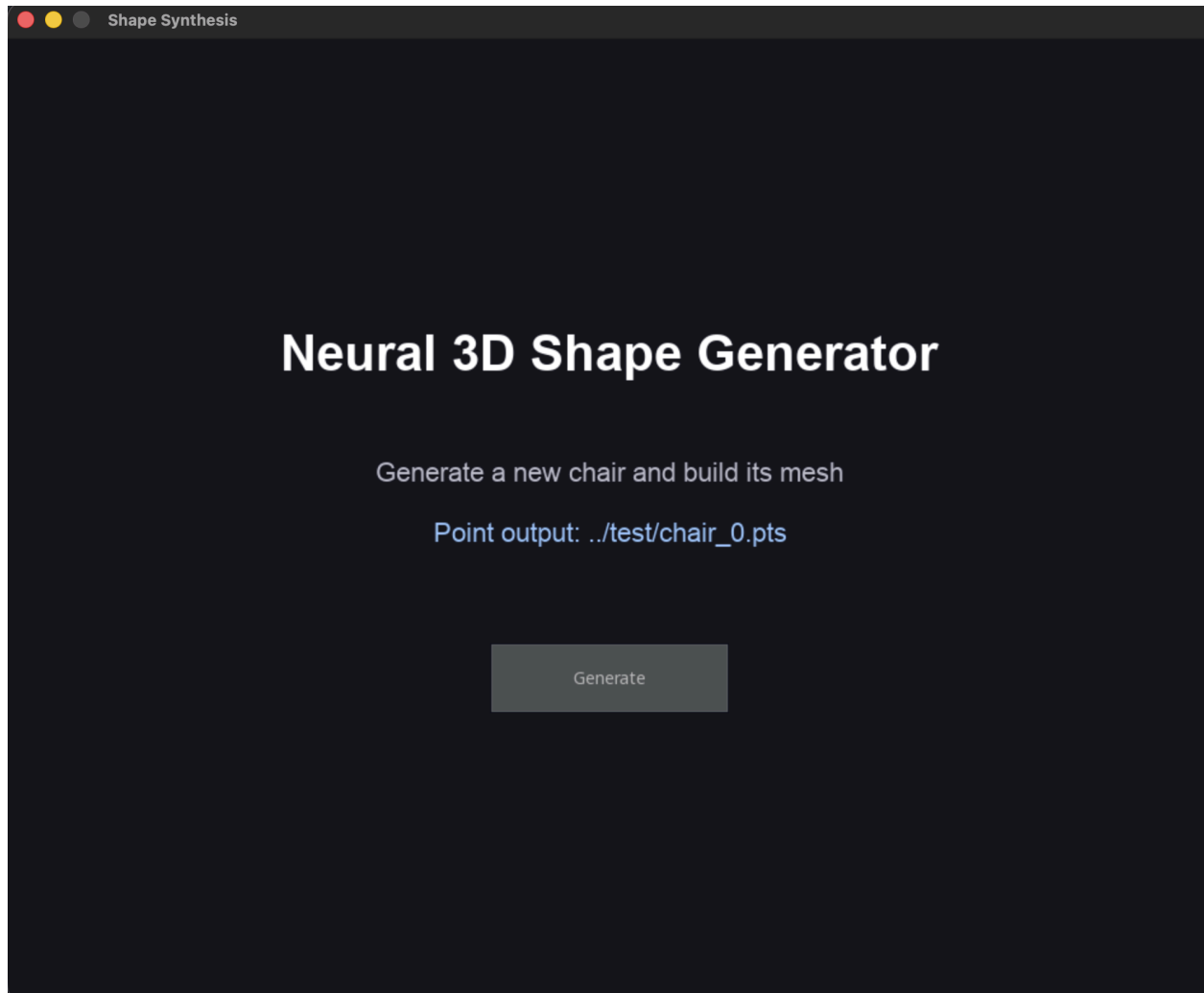


Figure 1: The landing page for the 3D chair generator.

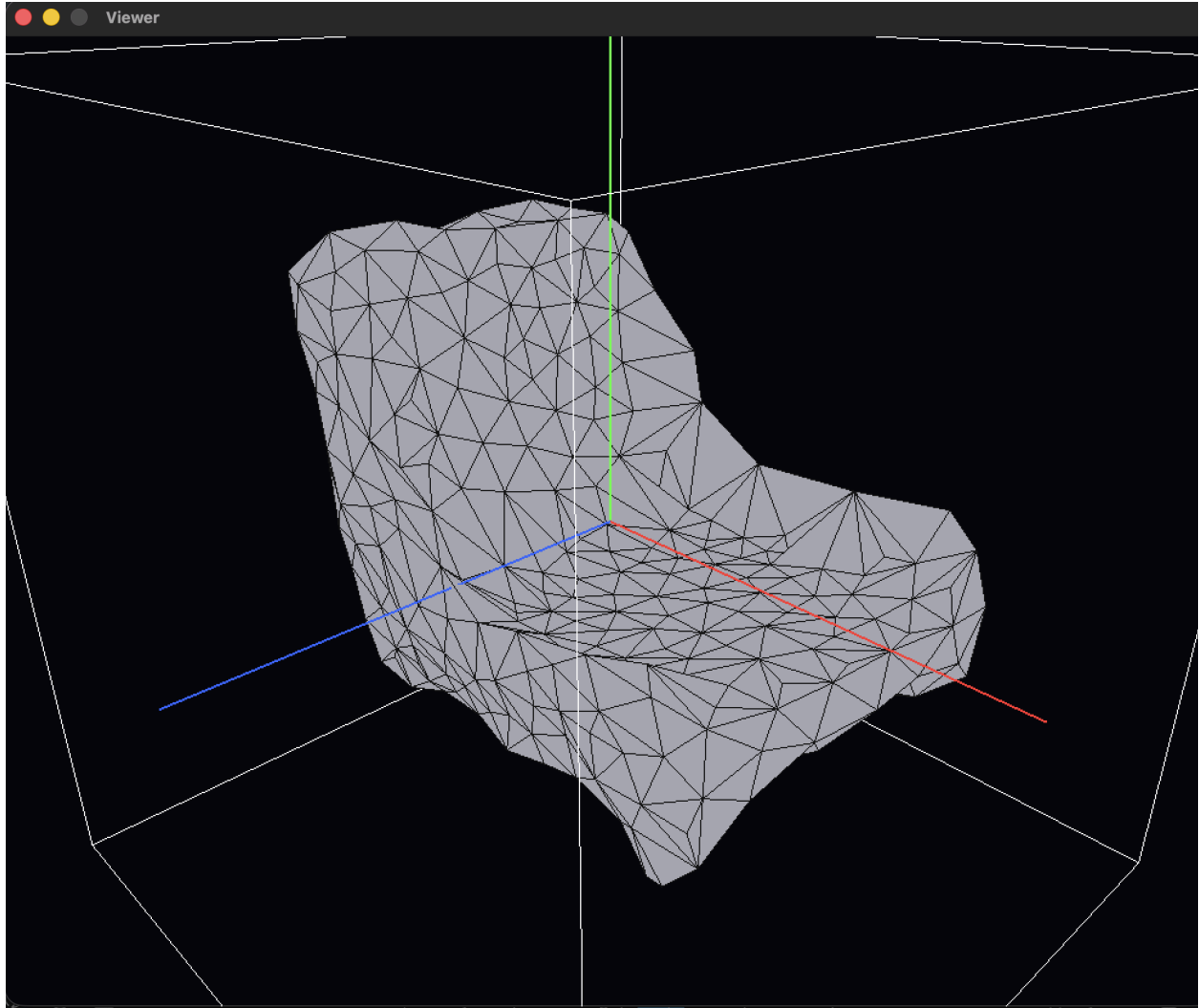


Figure 2: The interactive viewer showing a 3D-generated chair after the pipeline finishes.

Video & Code Links

- Project Video: [<https://youtu.be/WEoSTUP3NbY>]
- Code Repository: [<https://github.com/henryjlee729/shape-synthesis/tree/main>]
- Presentation:
[https://docs.google.com/presentation/d/1vDFJiwKKSyj_FSyrpk3ScDV_GR5EsrOZtXjrPpLMoTg/edit?usp=sharing]

Limitations & Future Work

One limitation of the current system is the quality of the generated point clouds, which can sometimes be noisy or incomplete. This affects the mesh reconstruction stage, where artifacts or irregular surfaces may appear. Additionally, the reconstruction process is also sensitive to parameter choices, requiring tuning to balance smoothness and detail. Another limitation is the generative model's capacity, which may struggle to capture fine geometric features or consistently produce high-quality outputs.

Future work could address these issues by using more advanced generative models, such as diffusion-based approaches, to improve shape quality. Expanding the system to support multiple object categories would increase its generality. On the graphics side, improvements such as enhanced shading, better surface reconstruction methods, or additional interaction features could further improve the visual quality of the 3D model.